Leukocyte Activation: The Mechanism

Step 1: Recognition

Receptors on the leukocyte detect the threat:

- **TLRs** (toll-like receptors) bind microbial products (e.g., TLR4 binds LPS, lipopolysaccharide)
- **Fc receptors and complement receptors** bind microbes coated with antibody or C3b (**opsonisation**)
- **GPCRs** (G-protein-coupled receptors) bind bacterial peptides and C5a
- **IFN-γ** (interferon-gamma) receptor on macrophages boosts activation

Step 2: Signalling

Receptor binding triggers signalling inside the cell (NF-κB, MAP kinases, **calcium rise**), which turns on genes and activates the killing machinery. From here the cell takes one of two routes, depending on the target.

Route A: Phagocytosis (small microbes)

1. The leukocyte binds the microbe and extends pseudopods around it.
2. It engulfs it, forming a **phagosome**.
3. The phagosome fuses with a **lysosome**, forming a **phagolysosome**.
4. **Killing inside the phagolysosome:**
    - **Respiratory burst:** NADPH oxidase converts O_2 → superoxide (O_2^-) → H_2O_2 (hydrogen peroxide). Myeloperoxidase (MPO) converts $H_2O_2 + Cl^-$ → **HOCl (bleach)**, the main killer.
    - **Nitric oxide:** macrophages make NO (via iNOS, inducible nitric oxide synthase), which forms peroxynitrite.
    - **Enzymes and peptides:** lysozyme, defensins, acid hydrolases.

Route B: NET formation (neutrophils, against large targets or when phagocytosis is overwhelmed)

NETs (neutrophil extracellular traps) are webs of DNA and granule proteins that trap and kill microbes **outside** the cell.

Mechanism (NETosis):

1. **Trigger:** strong neutrophil activation by microbes, LPS, immune complexes, IL-8 or activated platelets.
2. **ROS production:** NADPH oxidase generates reactive oxygen species. Calcium also rises inside the cell.
3. **Granule protein movement: MPO** and **neutrophil elastase** move from the granules into the nucleus.
4. **Citrullination of histones:** the calcium-dependent enzyme **PAD4** (peptidylarginine deiminase 4), located in the nucleus, converts **arginine → citrulline on the histones** (mainly H3 and H4).
    - Histones are normally **positively charged** and bind tightly to the **negatively charged DNA**.
    - Citrullination removes the positive charge, so histones **bind DNA weakly**.
    - Elastase and MPO also help break down histones.
5. **Chromatin decondensation:** the loosened DNA unwinds and swells.
6. **Rupture:** the nuclear envelope and cell membrane break, and the neutrophil dies (**NETosis**).
7. **Result:** a web of **decondensed chromatin (DNA + histones)** studded with **MPO, elastase, defensins** and other granule proteins.

How NETs kill:

- The web **physically traps** microbes and stops them spreading.
- Histones and granule proteins **kill** the trapped microbes directly.

Harm from NETs:

- **Autoimmunity:** NETosis exposes **citrullinated histones and proteins** and DNA to the immune system. These can act as autoantigens, a proposed source of **anti-CCP antibodies** (anti-cyclic citrullinated peptide) in **rheumatoid arthritis** and of anti-DNA/anti-histone antibodies in **SLE** (systemic lupus erythematosus).
- **Thrombosis:** NETs trap platelets and activate coagulation, contributing to **deep vein thrombosis**.

<u>Step 5: Release of mediators (both routes)</u>

Activated cells secrete cytokines (TNF, IL-1, IL-6), which amplify inflammation and recruit more leukocytes.

Step 6: Termination of the response

Activation must be switched off once the threat is gone, otherwise normal tissue is damaged. This happens through several linked mechanisms:

1. **Short life of the mediators and cells**
    - Most inflammatory mediators are **quickly broken down** or decay once the stimulus is removed (e.g., C5a and C3a are inactivated by carboxypeptidase N; arachidonic acid metabolites are short-lived).
    - **Neutrophils are short-lived** (hours to a few days) and die by **apoptosis** after leaving the blood, so the response fades unless new cells are recruited.
2. **Clearance of dead cells and microbes**
    - **Macrophages** engulf apoptotic neutrophils (**efferocytosis**), and also clear NET debris. NETs are broken down by **DNase** enzymes in the tissue and plasma. Failure to clear NETs and apoptotic debris promotes autoimmunity.
3. **Switch to anti-inflammatory signals**
    - Macrophages shift from pro-inflammatory (M1) to **anti-inflammatory / repair (M2)** behaviour after ingesting apoptotic cells.
    - They release **IL-10** (interleukin-10) and **TGF-β** (transforming growth factor beta), which suppress further activation.
    - **Specialised pro-resolving mediators: lipoxins, resolvins, protectins** (made from arachidonic acid and omega-3 fatty acids). They **stop neutrophil entry**, reduce chemokine production and promote clearance of dead cells. The switch from leukotrienes and prostaglandins (pro-inflammatory) to **lipoxins** is a key turning point.
4. **Negative feedback inside the cell**
    - Signalling is dampened by inhibitory proteins (e.g., **SOCS**, suppressors of cytokine signalling; **IκB** re-binds NF-κB; phosphatases switch off kinases).
    - **Inhibitory receptors** and **anti-inflammatory cytokines** also reduce activation.
5. **Outcome**
    - **Resolution and healing** if the injury is cleared: macrophages and growth factors drive repair.
    - **Chronic inflammation** if termination fails or the stimulus persists, with continued tissue damage.

One-line flow

Recognition → signalling (↑ calcium) → **either** phagocytosis → killing (ROS, NO, enzymes) **or** NETosis (ROS → MPO/elastase to nucleus → **PAD4: histone arginine → citrulline** → chromatin decondenses → cell ruptures → NET released) → cytokine release → **termination** (mediators decay, neutrophil apoptosis, efferocytosis and DNase clearance, IL-10/TGF-β and lipoxins, M1 → M2 switch) → resolution or chronic inflammation

Memory line: neutrophils kill with **HOCl** inside the cell and **NETs** outside it; macrophages add **NO**; **PAD4 citrullinates histones** to unwind DNA for NETs; **lipoxins and efferocytosis** switch the response off.